

Time: 1 Hrs.

Total Marks: 40

Note: Marks are indicated against each question.

Question No. 1

(8)

If P, Q and R are three atomic variables, obtain the Principal Disjunctive Normal form and Principal Conjunctive Normal form for the given formula without using TT.

$$(P \rightarrow (Q \wedge R)) \vee (\sim P \rightarrow (Q \vee R))$$

Question no. 2

(8)

Determine if the following inference pattern is Valid OR Invalid. Produce evidence (write the steps) to confirm its validity/invalidity.

$$\frac{(p \vee q) \rightarrow (s \wedge t) \quad [\sim((\sim s) \vee (\sim t))] \rightarrow [(\sim r) \vee q]}{(p \vee q) \rightarrow (r \rightarrow q)}$$

Question no. 3

(2+6)

What is a partition on a set? Let $A = \{1, 2, 3, 4, 5\}$ and a relation R defined on A is $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4), (4, 5), (5, 4), (5, 5)\}$. Obtain the cells of the partition.

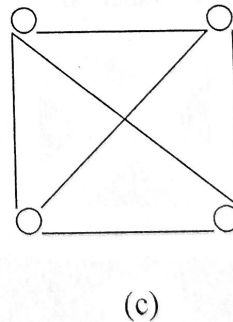
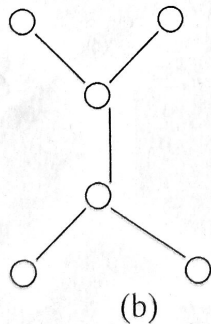
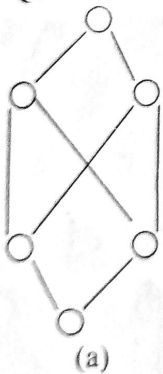
Question no. 4

(1+3+4)

What is a Hasse diagram? Draw the Hasse diagrams of the following sets under the partial ordering relation "divides" and indicate those which are totally ordered. Also, determine least member, greatest member, minimal member and maximal member in each case.

- (i) $\{2, 6, 24\}$
- (ii) $\{1, 2, 3, 6, 12\}$
- (iii) $\{3, 9, 27, 54\}$

Question no. 5 What is a lattice? Determine (show steps), which of the following graphs are lattice? (2+6)



School of Computer & Systems Science
CS-404: Discrete Mathematics
Mid-Semester Exam

Date: 10/11/2025
Time: 9:30 – 11:00am

Max Marks: 25

ANSWER QUESTIONS OF **MAXIMUM MARKS 25.**

- 1) Give the set of logical constants. Give one each and describe the well-formed formula for propositional logic and predicate calculus. [3]
- 2) Let C be a subset of a universal set $U = \{u_1, u_2, u_3, \dots, u_n\}$. Name, describe and state application of the function f_A defined as [2]

$$f_A(u_i) = \begin{cases} 1 & \text{if } u_i \in A \\ 0 & \text{if } u_i \notin A \end{cases}$$

- 3) In a given counting problem, how can you decide the use of permutation or combination? A Hackathon2026 must have **3 PhDs** and **5 MTech** students. Given **24 PhD** and **20 M.Tech.** eligible applicants, how many Hackathon2026 teams are possible? [3]
- 4) Let $A = \{a, b, c, d\}$, $B = \{1, 2, 3\}$ and $R_1 = \{(a, 2), (b, 1), (c, 2), (d, 1)\}$. Is R_1 a Function? Is R_1^{-1} a function? Justify your answer. [2]
- 5) Given a Boolean alphabet $B = \{0, 1\}$, define a *regular expression* considering the set of standard operators $\{-, +, *\}$. What is the length of the shortest string represented by the $(1 \circ 0)^*(1 \circ 0 + 1^*) (0^* \circ 1)$, justifying your answer. [3+2]
- 6) Differentiate *Relation* from *Function*. Let $A = \mathbb{R}$ and let $F = \{(a, 2a+1) \mid \forall a \in A\}$. What are the properties of F ? Is F a relation or a function? Justify your answer. [5]
- 7) Let $D = \{1, 2, 3, 4, 5, 6\}$ and for given cycles $p_1 = (5, 1, 4)$ and $p_2 = (2, 6, 3)$ complete the following and name the type of function these are. Give $p_1 \circ p_2$.

a) $(5, 1, 4) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ - & - & - & - & - & - \end{pmatrix}$

b) $(2, 6, 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ - & - & - & - & - & - \end{pmatrix}$

[5]

8) Consider the following and name the basic operators used to compute the set S and matrix $M_1 \odot M_2$. Give the total number of times the basic operators will be executed to compute S and $M_1 \odot M_2$.

i) Let $U = \{a, b, c, d, e, f, g, h, i, j\}$, $G = \{a, b, d, f\}$, $H = \{a, b, c, h, j\}$, $I = \{g, h\}$. To compute, $S = (\overline{G} \cap H) \cup I$.

ii) Let $M_1 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $M_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$. Let $\odot$ be the operator

Boolean product $A \odot B$ for $m \times n$ Boolean matrices say A , $m \times p$ matrix and B , $p \times n$ matrix, be denoted by $C = [c_{ij}]$ and defined by,

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ik} = 1 \text{ and } b_{kj} = 1 \text{ for some } k, \\ 0 & \text{otherwise} \end{cases} \quad 1 \leq k \leq p$$

[3+5]

9) Let $R \subseteq \mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) R (c, d)$ if and only if $a+d = b+c$. Give the properties of R . Give three applications of directed graphs in Computer Science.

[2+3]

10) Differentiate the **transitive closure** of an arbitrary set S from an arbitrary relation R . Give the transitive closure of $R_2 = \{(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 2), (3, 4), (4, 1)\}$.

[3]

11) Give full account of a **Hasse diagram**. Is lattice related to Hasse diagram? Justify your answer.

[5]